Dreissena rostriformis bugensis
(quagga mussel)
Mollusks-Bivalves
Exotic

Collection Info
Point Map
Species Profile
Animated Map
Impacts

Image 1 of 3

Profunda morph

Mike Quigley, NOAA ©

Dreissena rostriformis bugensis Andrusov, 1897

Common name: quagga mussel

Synonyms and Other Names: *Dreissena rostriformis bugensis* is viewed as a freshwater subspecies (or race) of *D. rostriformis* (Therriault et al. 2004).

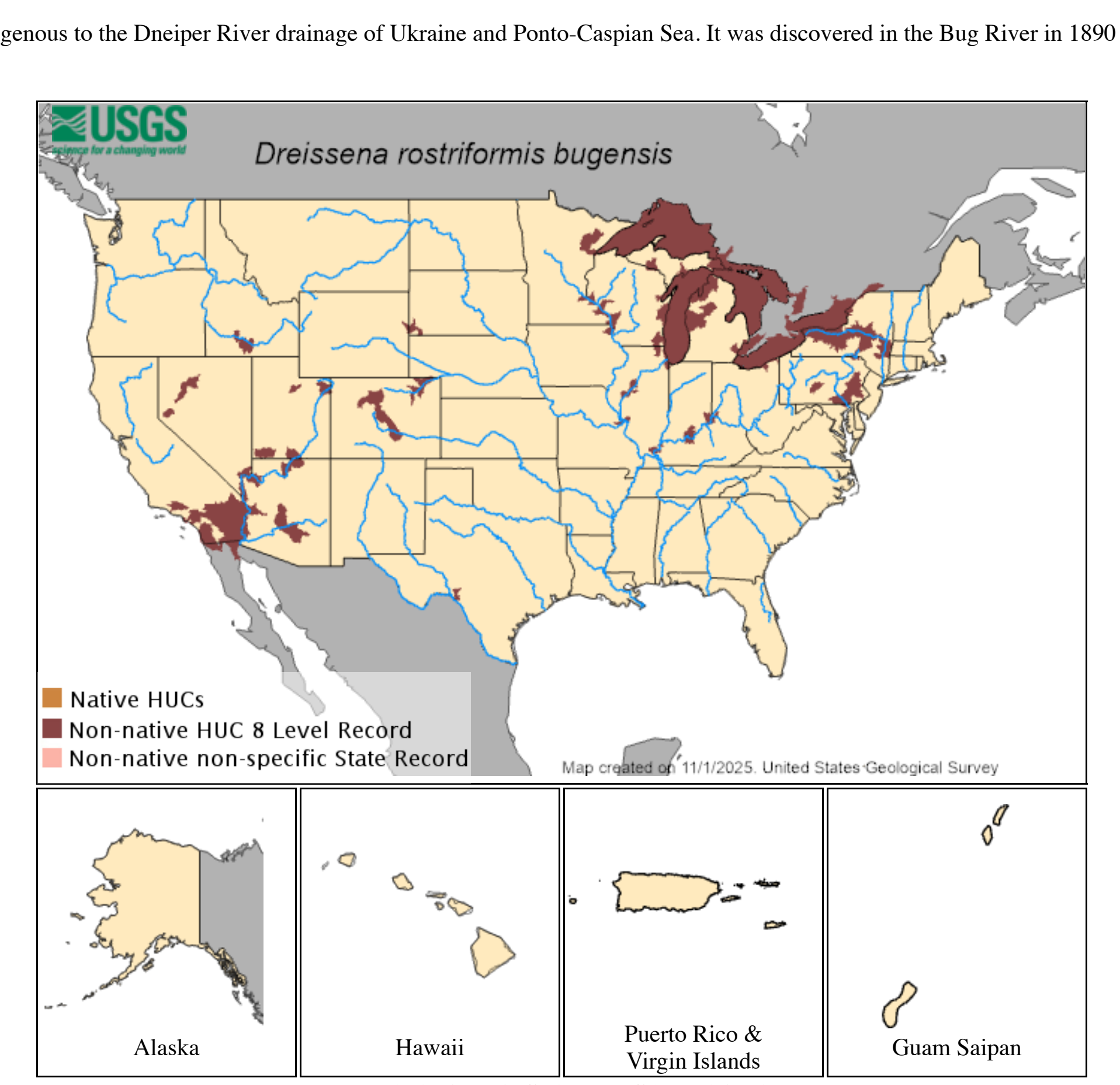
Taxonomy: available through ITIS

Injurious: This species is listed by the U.S. Fish and Wildlife Service as injurious wildlife.

Identification: *Dreissena rostriformis bugensis* is a small freshwater bivalve mollusk that exhibits many different morphs, though there are several diagnostic features that aid in identification. The quagga mussel has a rounded angle, or carina, between the ventral and dorsal surfaces (May and Marsden 1992). The quagga also has a convex ventral side that can sometimes be distinguished by placing shells on their ventral side: a quagga mussel will topple over, whereas a zebra mussel will not (Claudi and Mackie 1994). Overall, quaggas are rounder in shape and have a small byssal groove on the ventral side near the hinge (Claudi and Mackie 1994). Color patterns vary widely with black, cream, or white bands; a distinct quagga morph has been found that is pale or completely white in Lake Erie (Marsden et al. 1996). They usually have dark concentric rings on the shell and are paler in color near the hinge. If quaggas are viewed from the front or from the ventral side, the valves are clearly asymmetrical (Domm et al. 1993). Considerable phenotypic plasticity of all morphological characteristics is known in dreissenid species and this may be a result of environmental factors, meaning the same genotype may express different phenotypes in response to environmental conditions (Claxton et al. 1998). Due to this phenotypic plasticity, visual identification is not always an acceptable means of differentiating between quagga and zebra mussels (Kerambrun et al. 2018, Beggel et al. 2015). Thus, different methods of genetic comparison have been developed (e.g. May and Marsden 1992; Brown and Stepien 2010; Ram et al. 2012).

Size: Reaching sizes up to 4 cm

Native Range: *Dreissena rostriformis bugensis* is indigenous to the Dneiper River drainage of Ukraine and Ponto-Caspian Sea. It was discovered in the Bug River in 1890 by Andrusov, who named the species in 1897 (Mills et al. 1996).


USGS
science for a changing world
Dreissena rostriformis bugensis
Native HUCs
Non-native HUC 8 Level Record
Non-native non-specific State Record
Map created on 11/1/2025. United States Geological Survey
Alaska
Hawaii
Puerto Rico & Virgin Islands
Guam Saipan

Hydrologic Unit Codes (HUCs) Explained
Interactive maps: Point Distribution Maps

Nonindigenous Occurrences:

Table 1. States with nonindigenous occurrences, the earliest and latest observations in each state, and the tally and names of HUCs with observations†. Names and dates are hyperlinked to their relevant specimen records. The list of references for all nonindigenous occurrences of *Dreissena rostriformis bugensis* are found here.

State	First Observed	Last Observed	Total HUCs with observations†	HUCs with observations†
AZ	2007	2021	10	Agua Fria; Bill Williams; Havasu-Mohave Lakes; Imperial Reservoir; Lake Mead; Lower Colorado; Lower Colorado-Marble Canyon; Lower Lake Powell; Lower Salt; Middle Gila
CA	2007	2025	12	Havasu-Mohave Lakes; Imperial Reservoir; Newport Bay; Salton Sea; San Diego; San Gabriel; San Luis Rey-Escondido; Santa Ana; Santa Clara; Santa Margarita; Southern Mojave; Whitewater River
CO	2007	2017	5	Blue; Colorado Headwaters; Middle South Platte-Sterling; South Platte Headwaters; Upper Arkansas
ID	2023	2024	1	Upper Snake-Rock
IL	2002	2023	5	Chicago; Lake Michigan; Lower Illinois-Lake Chautauqua; Lower Ohio-Bay; Pike-Root
IN	2003	2014	1	Lake Michigan
IA	2006	2006	1	Coon-Yellow
KY	2004	2005	3	Blue-Sinking; Lower Ohio-Bay; Middle Ohio-Laughery
MI	1997	2023	13	Betsie-Platte; Boardman-Charlevoix; Brule; Carp-Pine; Cheboygan; Detroit; Fishdam-Sturgeon; Lake Erie; Lake Huron; Lake Michigan; Lake St. Clair; Muskegon; Pere Marquette-White
MN	2004	2006	5	Buffalo-Whitewater; La Crosse-Pine; Lake Superior; Rush-Vermillion; St. Louis
MO	1995	1995	1	Peruque-Piasa
NV	2007	2023	4	Havasu-Mohave Lakes; Lake Mead; Lower Humboldt; Middle Carson
NY	1991	2024	14	Chaumont-Perch; Headwaters St. Lawrence River; Irondequoit-Ninemile; Lake Erie; Lake Ontario; Lower Genesee; Middle Hudson; Mohawk; Niagara River; Oak Orchard-Twelvemile; Oneida; Raisin River-St. Lawrence River; Seneca; Upper Susquehanna
OH	1992	2023	3	Ashtabula-Chagrin; Lake Erie; Middle Ohio-Laughery
PA	1994	2013	5	Lake Erie; Lehigh; Lower Susquehanna; Schuylkill; Upper Juniata
SD	2014	2014	1	Angostura Reservoir
TX	2021	2022	1	Lower Devils
UT	2009	2014	3	Lower Green-Diamond; Provo; Upper Virgin
WI	2000	2024	8	Buffalo-Whitewater; Coon-Yellow; Duck-Pensaukee; Lake Michigan; Lake Superior; Manitowoc-Sheboygan; Rush-Vermillion; Upper Fox

Table last updated 12/3/2025

† Populations may not be currently present.

Ecology: Quagga mussels inhabit freshwater rivers, lakes, and reservoirs. In North American populations, they are not known to tolerate salinities greater than 5 ppt (Spidle et al. 1995). Water temperatures of 28°C begin to cause significant mortality, and 32-35°C are considered lethal for dreissenid species (Antonov and Shkorbatov 1990, as cited in Mills 1996). The depth at which the mussels live varies depending on water temperature. They are not generally found in lakes near shore in shallow water due to wave action. The quagga mussel can inhabit both hard and soft substrates, including sand and mud, down to depths of 130 m and possibly deeper. The maximum density of quagga mussels in Lake Michigan is at 31-90 m (Rowe et al. 2015a).

Means of Introduction: The introduction of *D. r. bugensis* into the Great Lakes appears to be the result of ballast water discharge from transoceanic ships that were carrying veligers, juveniles, or adult mussels. The genus *Dreissena* is highly polymorphic and prolific, with high potential for rapid adaptation attributed to its rapid expansion and colonization (Mills et al. 1996). Still, there are other factors that can aid in the spread of this species across North American waters. Thse factors include larval drift in river systems or fishing and boating activities that allow for overland transport or movement between water basins.

Status: The quagga mussel may have arrived more recently than the zebra based on differences in size classes of initially discovered populations, and therefore it seems plausible that the quagga is still in the process of expanding its nonindigenous range (May and Marsden 1992, MacIsaac 1994). In the 1990s, the absence of quagga mussels from areas where zebra mussels were present may have been related to the timing and location of introduction rather than physiological tolerances (MacIsaac 1994). The quagga mussel is now well established in the lower Great Lakes and found in a few harbor and nearshore areas of ake Superior.

Quagga mussels have displaced zebra mussels in all offshore areas of Lakes Michigan (Nalepa et al 2014); Rowe et al. 2015a), Huron (Nalepa et al. 2018), and Ontario (Wilson et al. 2006; Birkett et al. 2015). There is a gradient of dreissenid domination in Lake Erie, with quagga mussels dominating eastern basins and the two species coexisting in the western basin (Patterson et al. 2005; Karatayev et al. 2014). A similar gradient was initially observed in southern Lake Ontario with quagga mussel dominating the west and zebra dominating the east (Mills et al. 1999), but the quagga mussel has since displaced zebra mussels in all offshore regions of Lake Ontario (Birkett et al. 2015). Coexistence is generally only found in shallow, productive systems such as Green Bay in Lake Michigan, Saginaw Bay in Lake Huron, and Western Lake Erie.There are multiple mechanisms by which quagga mussels displace zebra mussels, including differences in growth, respiration, and development (Ram et al. 2012; Karateyev et al. 2015). Though zebra mussels have garnered the majority of public and research attention, quagga mussels have a more extensive distribution in the Great Lakes and their abundance far exceeds that of the zebra mussel peak (e.g., southern Lake Michigan, Nalepa et al. 2010).

Impact of Introduction:

Summary of species impacts derived from literature review. Click on an icon to find out more...

Ecological	Economic	Human Health	Other

Quaggas are prodigious water filterers, removing substantial amounts of phytoplankton and suspended particulate from the water. As such, their impacts are similar to those of the zebra mussel. By removing the phytoplankton, quaggas in turn decrease the food source for zooplankton, therefore altering the food web. Impacts associated with the filtration of water include increases in water transparency, decreases in mean chlorophyll a concentrations, and accumulation of pseudofeces (Claxton et al. 1998). Water clarity increases light penetration causing a proliferation of aquatic plants that can change species dominance and alter the entire ecosystem. The pseudofeces that is produced from filtering the water accumulates and creates a foul environment. As the waste particles decompose, oxygen is used up, and the pH becomes very acidic and toxic byproducts are produced. In addition, quagga mussels accumulate organic pollutants within their tissues to levels more than 300,000 times greater than concentrations in the environment and these pollutants are found in their pseudofeces, which can be passed up the food chain, therefore increasing wildlife exposure to organic pollutants (Snyder et al. 1997). Macksasitorn et al. (2015) found that mussel tissue polychlorinated biphenyl (PCB) concentration was positively related to sediment PCB levels, suggesting that quagga (and zebra) mussels might provide an entry point for PCBs into near-shore benthic trophic webs.

Dreissena species ability to rapidly colonize hard surfaces causes serious economic problems. These major biofouling organisms can clog water intake structures, such as pipes and screens, therefore reducing pumping capabilities for power and water treatment plants, cooling industries, companies, and communities. Recreation-based industries and activities have also been impacted; docks, breakwalls, buoys, boats, and beaches have all been heavily colonized. Quaggas are able to colonize both hard and soft substrata so their negative impacts on native freshwater mussels, invertebrates, industries and recreation are unclear. Many of the potential impacts of *Dreissena* are unclear due to the limited time scale of North American colonization. Nonetheless, it is clear that the genus *Dreissena* is highly polymorphic and has a high potential for rapid adaptation to extreme environmental conditions by the evolution of allelic frequencies and combinations, possibly leading to significant long-term impacts on North American waters (Mills et al. 1996). *Dreissena rostriformis bugensis* lacks the keeled shape that allows *D. polymorpha* to attach so tenaciously to hard substrata; though, *D. rostriformis bugensis* is able to colonize hard and soft substrata (Mills et al. 1996). The ability to colonize different substratas could suggest that *D. rostriformis bugensis* is not limited to deeper water habitats and that it may inhabit a wider range of water depths where they have been found at depths up to 130 m in the Great Lakes (Mills et al. 1996, Claxton and Mackie 1998).

Remarks: Hybridization between the two introduced dreissenid species was an initial concern. Zebra x quagga mussel hybrids were created by pooling gametes collected after exposure to serotonin in the laboratory, indicating that interspecies fertilization may be feasible (Mills et al. 1996). However, there is evidence for species-specific sperm attractants suggesting that interspecific fertilization may be rare in nature. Thus, if hybridization does occur, these hybrids will constitute a very small proportion of the dreissenid community (Mills et al. 1996). There is evidence to suggest that the apparent ability of *Dreissena rostriformis bugensis* to outcompete *Dreissena polymorpha* is due to the high mtility of the species (D'Hont et al. 2021).

Redear sunfish (*Lepomis microlophus*) have been shown in experimental enclosers in Sweetwater Reservoir, CA to feed upon and control population sizes of quagga mussels (Wong et al. 2013).

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Other Resources:
Current Printable Zebra and Quagga Mussel Sightings Distribution Map (PNG) (PDF)

Quagga Mussel Images (Public Domain)

USGS *Dreissena* species Frequently Asked Questions

USGS Zebra Mussel Frequently Asked Questions

Invasive Mussel Collaborative

Dreissena Project Coordination Mapper

Student Questions Concerning Zebra and Quagga Mussels

USGS Zebra Mussel Fact Sheet

Great Lakes Waterlife

US Fish and Wildlife Service Ecological Risk Screening Summary for *Dreissena rostriformis*

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